HAALSA Hackathon — results brief

Sat Oct 24, 10 AM–5 PM ET

Zoom: `https://bostonu.zoom.us/j/96853036635` ID 968 5303 6635 passcode 421911.
Folder `research/haalsi-hackathon/`: untouched starter `HAALSI_hackathon_student.Rmd`; filled copy with all answers `HAALSI_hackathon_joshua.Rmd` / `.html`.

Pitch: Question 1 — which intervention could cut cognitive impairment most?

Data. HAALSI, rural Agincourt, South Africa: 5,059 adults aged 40+ at Wave 1, three waves over about 6 years (12,092 person-waves with cognition). **Outcome**: `objective_cognition_z` (orientation + immediate + delayed word recall, Wave 1 SD units) and `low_cognition` (below the Wave 1 20th percentile). **Confounders**: age, sex, education (all come before the exposures). Wealth is added for hypertension and depression. BMI, hypertension and depression are *not* adjusted for in the wealth model because they may be mechanisms. **Methods**: linear regression (level), logistic regression (low cognition), linear mixed model with random intercept and slope (change), then predicted prevalence under each intervention.

Exposure	Level (SD)	OR low cognition	Modelled change in low cognition
Wealth, per 1 SD assets	+0.11 (0.08, 0.14)	0.70 (0.63, 0.78)	+0.5 SD assets: 19.9% → 17.8% (−2.2 pts)
Hypertension, untreated	+0.01 (−0.04, 0.07)	1.01 (0.84, 1.23)	
Hypertension, treated	+0.16 (0.10, 0.22)	0.61 (0.50, 0.76)	treat all untreated: −2.1 pts
Depression, per 1 SD CES-D	−0.11 (−0.13, −0.08)	1.26 (1.17, 1.36)	−0.5 SD if symptomatic: −0.6 pts

How to read it.

- **Wealth**: graded — richest vs poorest quintile +0.31 SD. A 0.5 SD rise in assets removes about 11% of low-cognition cases. It does *not* change the 6-year slope (0.002 SD/yr).
- **Hypertension**: untreated looks the same as none. The whole "benefit" is treated vs untreated people, who differ in clinic access, health literacy and resources (healthy-user bias). This is the estimate we trust least.
- **Depression**: linked to self-reported difficulty almost twice as strongly (0.17–0.20 SD) as to test scores (−0.11). Self-reports over-flag depressed people.

Verdict: financial security (cash transfer / Old Age Pension) is the most promising. It has the largest modelled effect, the most credible mechanism and existing natural-experiment evidence (Case 2001; Duflo 2003; Ardington et al. 2009), and it benefits the whole household.

Limitations (pick 1–2): (1) confounding by lifelong factors — childhood circumstances and cognitive ability shape both wealth and scores. (2) Healthy-user bias in the treated-hypertension contrast. (3) Practice effects and dropout or death (about 1 in 5 by Wave 3): scores *rise* 0.08 SD/yr on retest, so "decline" analyses compare gains. (4) A short cognitive test with ceiling effects.

Implication / better study: a regression discontinuity at pension-eligibility age 60, or a randomized cash transfer with cognitive follow-up.

Backup findings (if the team picks another question)

- **Q2 self-rated memory**: Spearman $\rho = -0.38$. "Fair/poor" catches only 37% of low-cognition cases (specificity 0.86, PPV 0.40). Related, but a poor screen.
- **Q3 diabetes**: objective cognition no different (−0.07, CI crosses 0), but worse self-rated memory (+0.13 SD). Only 249 treated — imprecise.
- **Q4 BMI**: underweight −0.20 SD, obese +0.20 SD, non-linear (spline $p < 0.001$). Likely reverse causation (illness or poverty causes thinness). No moderation by hypertension ($p = 0.45$).
- **Q5 ADLs**: hypertension *and* diabetes together ≈ doubles odds of any ADL difficulty (OR 2.02, 1.42–2.85). Either alone is not clearly raised. Interaction underpowered.
- **Q6/Q7 change**: education and older age predict *smaller practice gains* (a ceiling and regression-to-the-mean artefact, not harm). In the residual approach, wealth predicts outperforming one's expected score (OR 1.25/SD), but not in the slope model — the two approaches disagree.

Slide outline (5–6 slides): question & why it matters → data + variables + why those confounders → methods → the table above → verdict + limitations → better study. Required citation on the final slide: *HAALSI is sponsored by the National Institute on Aging (5P01AG041710-05) and conducted by the Harvard Center for Population and Development Studies in partnership with Wits University.*